Question # 4 of 5 ( Start time: 02:57:02 PM, 24 July 2023 )

Investigate the limit of $\lim_{x \to 0^-} \left(\frac{|x|}{x} + x\right) = ----$.

**Select the correct option**

| | |
|---|---|
| ○ | $-1$ |
| ○ | $0$ |
| ○ | *none of these* |
| ○ | $1$ |

Network
Recycle Bin
Control Panel
Dolby Audio

Network

Recycle Bin

Control Panel

Dolby Audio

**Question # 3 of 5 ( Start time: 02:56:17 PM, 24 July 2023 )**

The function $g(x) = x^2$ is

**Select the correct option**

- decreasing on $[0, \infty)$.
- having negative values on $[0, \infty)$.
- increasing on $[0, \infty)$.
- undefined on $[0, \infty)$.

Question # 1 of 5 ( Start time: 02:54:52 PM, 24 July 2023 )

The function $g(x) = -x^3$ is decreasing on ----.

Select the correct option

- $(-\infty, \infty)$
- $(-\infty, 0)$
- none of these
- $[0, \infty)$

Type here to search

4G 4G 2:49 47
vulms.vu.edu.pk/Quiz/Q 13

BC230209270: ANEEZA AZIZ

Time Left 81 sec(s)

MTH621 - Real Analysis I (Quiz # 3)

Quiz Start Time: 02:46 PM, 24 July 2023

Question # 5 of 5 ( Start time: 02:49:05 PM, 24 July 2023 )

Total Marks: 1

The function $g(x) = -x^3$ is decreasing on $-----$.

Select the correct option

Reload Math Equations

- $[0, \infty)$
- none of these
- $(-\infty, 0)$
- $(-\infty, \infty)$

Click to Save Answer & Move to Next Question

4G 4G 2:46 4G2 48
vulms.vu.edu.pk/Quiz/Q 13

The function $h(x) = -x^3$ is

**elect the correct option**

- positive on $(-\infty, \infty)$.
- increasing on $(-\infty, \infty)$.
- is undefined on $(-\infty, \infty)$.
- decreasing on $(-\infty, \infty)$.

Click to Sa

4G 4G 2:44 49
vulms.vu.edu.pk/Quiz/Q 13

*Investigate the* $\lim it\ of\ \lim_{x \to 0} 2x \sin \sqrt{x} = - - - -.$

**Select the correct option**

- $x$
- *none of these*
- 0
- 1

Click to Save Ans

Question # 2 of 5 ( Start time: 02:43:39 PM, 24 July 2023 ) Total Marks: 1

*Investigate the* $\lim$ *it of* $\lim_{x \to 0} \frac{\sin x}{x} = - - - - -$.

Select the correct option

Reload Math Equations

- $x$
- *none of these*
- 0
- 1

Click to Save Answer & Move to Next Question

4G 4G 2:43
4G 49
vulms.vu.edu.pk/Quiz/Q
13

Question # 1 of 5 ( Start time: 02:42:52 PM, 24 July 2023 )

*Investigate the limit of* $\lim_{x \to 0^-} \frac{1}{x} = - - - -.$

Select the correct option

| | |
|---|---|
| ○ | $\infty$ |
| ○ | $0$ |
| ○ | *none of these* |
| ○ | $-\infty$ |

Click to Save Answer

MTH621 - Real Analysis I (Quiz # 3) Quiz Start Time: 02:20 PM, 24 July 2023

Question # 5 of 5 ( Start time: 02:26:06 PM, 24 July 2023 ) Total Marks: 1

If $f : I \to \mathbb{R}$ has a derivative at $c \in I$, then $f$ is

Select the correct option

Reload Math Equations

- continuous at c.
- having one sided limit.
- discontinuous at c.
- undefined at c.

Click to Save Answer & Move to Next Question

Question # 4 of 5 ( Start time: 02:25:19 PM, 24 July 2023 ) Total Marks: 1

*Investigate the* $\lim it\ of\ \lim_{x\to 0} x\sin\frac{1}{x} = ----.$

Select the correct option

Reload Math Equations

| | |
|---|---|
| ○ | 1 |
| ○ | $x$ |
| ○ | *none of these* |
| ○ | 0 |

Click to Save Answer & Move to Next Question

**Question # 3 of 5 ( Start time: 02:24:23 PM, 24 July 2023 )** **Total Marks: 1**

*Investigate the* $\lim it\ of\ \lim_{x\to 0} \frac{\sin x}{x} = - - - -.$

**Select the correct option**

Reload Math Equations

| | |
|---|---|
| ○ | *none of these* |
| ○ | 0 |
| ○ | $x$ |
| ○ | 1 |

Click to Save Answer & Move to Next Question

2:23 59%
Quiz
https://vulms.vu.edu.pk/Quiz/QuizQuestio

**MTH621 - Real Analysis I (Quiz # 3)**

**Question # 5 of 5 ( Start time: 02:23:33 PM, 24 July 2023 )**

*Investigate the limit of* $\lim_{x \to 0^-} \left(\frac{|x|}{x} + x\right) = ----.$

**Select the correct option**

| | |
|---|---|
| ○ | *none of these* |
| ○ | 1 |
| ○ | 0 |
| ○ | $-1$ |

2:22 59%
Quiz
https://vulms.vu.edu.pk/Quiz/QuizQuestio

BC230206977: AYESHA ZAFAR

MTH621 - Real Analysis I (Quiz # 3)

**Question # 4 of 5 ( Start time: 02:22:54 PM, 24 July 2023 )**

The inverse of the function $f(x) = x^2$, is

**Select the correct option**

- $f^{-1}(y) = \sqrt{y}$.
- $f^{-1}(y) = \frac{1}{x^2}$
- $f^{-1}(y) = y^2$
- $f^{-1}(y) = \frac{1}{y^2}$

Question # 3 of 5 ( Start time: 02:22:13 PM, 24 July 2023 )

The function $g(x) = x^2$ is

**Select the correct option**

| | |
|---|---|
| ○ | undefined on $[0, \infty)$. |
| ○ | decreasing on $[0, \infty)$. |
| ○ | having negative values on $[0, \infty)$. |
| ○ | increasing on $[0, \infty)$. |

Question #1 of 5 ( Start time: 02:21:00 PM, 24 July 2023 ) Total Marks: 1

The function $f(x) = \begin{cases} x, & 0 \leq x < 1, \\ 2, & 1 \leq x \leq 2, \end{cases}$ is

Select the correct option

Reload Math Equations

- is nondecreasing on $I = [0, 2]$.
- is undefined on $I = [0, 2]$.
- is nonincreasing on $I = [0, 2]$.
- is having negative values on $I = [0, 2]$.

Click to Save Answer & Move to Next Question

2:21 59%
Quiz
https://vulms.vu.edu.pk/Quiz/QuizQuestio

(Quiz # 3) Quiz Start Time: 0

time: 02:21:06 PM, 24 July 2023 )

The function $g(x) = -x^3$ is decreasing on ------.

Reload

none of these

$(-\infty, \infty)$

$(-\infty, 0)$

$[0, \infty)$

Click to Save Answer & Move

$x \to x_0$

$$\lim_{x \to 0-} \frac{1}{x} = -\infty,$$

$$\lim_{x \to 0+} \frac{1}{x} = \infty;$$

$$\lim_{x \to 0-} \frac{1}{x^2} = \lim_{x \to 0+} \frac{1}{x^2} = \infty$$

$$\lim_{x \to 0} \frac{1}{x^2} = \infty;$$

$$\lim x^2 = \lim_{x \to -\infty} x^2 = \infty;$$

Example: Prov... $\lim_{x \to 0+} \sqrt{x} = 0$.

Solution: See Lecture.

Example: Show that

$$\lim_{x \to 0+} \left( \frac{|x|}{x} + x \right) = 1, \qquad \lim_{x \to 0-} \left( \frac{|x|}{x} + x \right) = -1.$$

Solution: See Lecture

Theorem: A function $f$ has a limit at $x_0$ if and only if it has left- and right-h... limits at $x_0$, and they are equal. More specifically,

$$f'(x_0) = \lim_{x \to x_0} \sum_{k=0} x^{n-k-1} x_0^k = n x_0^{n-1}.$$

ince this holds for every $x_0$, we drop the subscript and write

$$f'(x) = nx^{n-1} \quad \text{or} \quad \frac{d}{dx}(x^n) = nx^{n-1}.$$

ample: Find the derivative of the line $y = mx + c$.

lution:

Since $R_g$ = [illegible] theorem, we have proved with $f$ and [illegible] implies that $g$ is continuous on $[c, d]$.

**Example**: If

$$f(x) = x^2, \quad 0 \le x \le R.$$

**Solution**: The inverse of the given function is

$$f^{-1}(y) = g(y) = \sqrt{y}, \quad 0 \le y \le R^2.$$

**Example**: If

$$f(x) = 2x + 4, \quad 0 \le x \le 2,$$

**Solution**: The inverse of the given function is

$$f^{-1}(y) = g(y) = \frac{y-4}{2}, \quad 4 \le y \le 8.$$

In either case, $f$ is on $I$.

If $\leq$ can be replaced by $<$ in (3.12), $f$ is *increasing* on $I$. If $\geq$ can be replaced by $>$ in (3.13), $f$ is *decreasing* on $I$. In either of these two cases, $f$ is *strictly monotonic* on $I$.

**Example**: The function

$$f(x) = \begin{cases} x, & 0 \leq x < 1, \\ 2, & 1 \leq x \leq 2, \end{cases}$$

is nondecreasing on $I = [0, 2]$.

**Example**: The function $g(x) = x^2$ is increasing on $[0, \infty)$. The function $h(x) = -x^3$ is decreasing on $(-\infty, \infty)$.

**Theorem**: Suppose that $f$ is monotonic on $(a, b)$ and define

$\lim_{x \to 0+} \sqrt{x} = 0.$

Solution: See Lecture.

Example: Show that

$$\lim_{x \to 0+} \left( \frac{|x|}{x} + x \right) = 1, \qquad \lim_{x \to 0-} \left( \frac{|x|}{x} + x \right) = -1.$$

Solution: See Lecture

Theorem: A function $f$ has a limit at $x_0$ if and only if it has left- and limits at $x_0$, and they are equal. More specifically,

on $I$.

**Example**: The function

$$f(x) = \begin{cases} x, & 0 \le x < 1, \\ 2, & 1 \le x \le 2, \end{cases}$$

is nondecreasing on $I = [0, 2]$.

**Example**: The function $g(x) = x^2$ is increasing on $[0, \infty)$. The function is decreasing on $(-\infty, \infty)$.

**Theorem**: Suppose that $f$ is monotonic on $(a, b)$ and define

$\alpha = \inf f(x)$ and $\beta = \sup f(x)$.

**Example**: For the function

$$f(x) = x \sin \frac{1}{x}, \quad x \neq 0,$$

show that

$$\lim_{x \to 0} f(x) = 0$$

**Solution**: Even though $f$ is not defined at $x_0 = 0$, because if

$$0 < |x| < \delta = \varepsilon,$$

then

$$|f(x) - 0| = \left| x \sin \frac{1}{x} \right| \leq |x| < \varepsilon.$$

On the other hand, the function

$$g(x) = \sin \frac{1}{x}, \quad x \neq 0,$$

has no limit as $x$ approaches 0, since it assumes all values between $-1$ and 1 in every neighborhood of the origin.

creasing.

$R_g = \{g(y) : y \in [c, d]\}$ is the interval $[g(c), g(d)] = [a, b]$, t

we have proved with $f$ and $[a, b]$ replaced by $g$ and $[c, d]$ imp

s on $[c, d]$.

: If

$$f(x) = x^2, \quad 0 \le x \le R.$$

The inverse of the given function is

Solution

$$f^{-1}(y) = g(y) = \sqrt{y}, \quad 0 \le y \le R^2.$$

: If

$$f(x) = 2x + 4, \quad 0 \le x \le 2.$$

The inverse of the given function is

$$f^{-1}(y) = g(y) = \frac{y - 4}{2}, \quad 4 \le y \le 8.$$